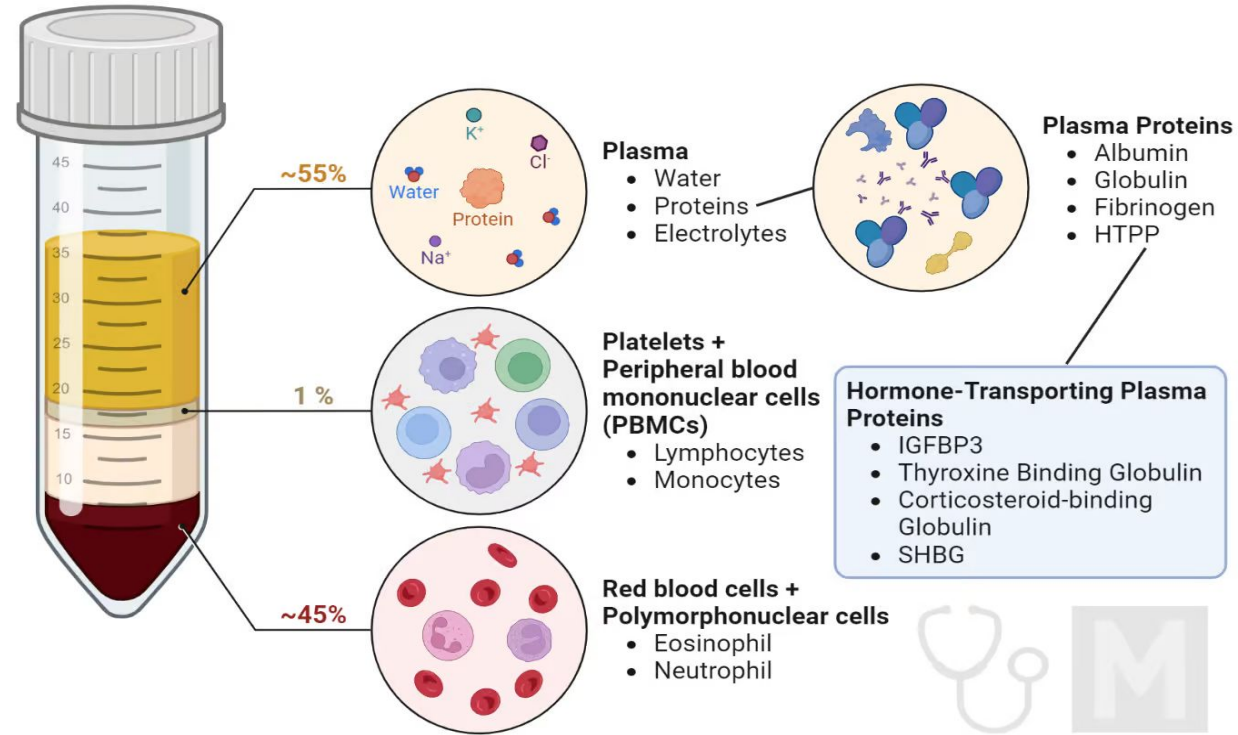


PLASMA PROTEINS



CLINICAL SIGNIFICANCE OF PLASMA PROTEIN AND IMMUNOGLOBULIN ABNORMALITIES

DR WAJEEHA SHAMS ANSARI

INTRODUCTION TO PLASMA PROTEINS

* **Definition:** Soluble proteins in blood plasma (Albumin, Globulins, Fibrinogen).

* **Functions:**

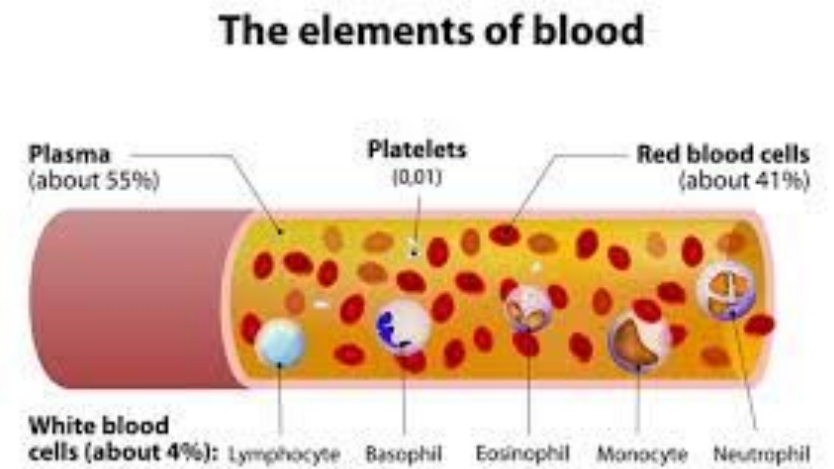
* Maintenance of colloid osmotic pressure(Albumin).

* Transport (Lipoproteins, Transferrin, Ceruloplasmin).

* Immunity (Immunoglobulins).

* Clotting (Fibrinogen).

* **Clinical Relevance:** Most are synthesized in the liver (except gamma-globulins). Alterations provide diagnostic clues for systemic disease.



ALBUMIN ABNORMALITIES

- * **Hypoalbuminemia (Low Albumin):**

- * **Causes:** Liver disease (decreased synthesis), malnutrition, kidney disease (nephrotic syndrome - loss in urine), burns.

- * **Clinical Sign:** Edema (due to low osmotic pressure).

- * **Clinical Link:** Poor wound healing and increased risk of postoperative edema in dental procedures.



ACUTE PHASE PROTEINS (APPS)

- * **Concept:** Proteins whose concentrations change by at least 25% during inflammation.
- * **Positive APPs:** CRP (C-Reactive Protein), Fibrinogen, Ferritin. (Increase during inflammation).
- * **Negative APPs:** Albumin, Transferrin. (Decrease during inflammation).
- * **Dental Context:** CRP is a marker of chronic systemic inflammation and is strongly correlated with the severity of Periodontitis

INTRODUCTION TO IMMUNOGLOBULINS (IG)

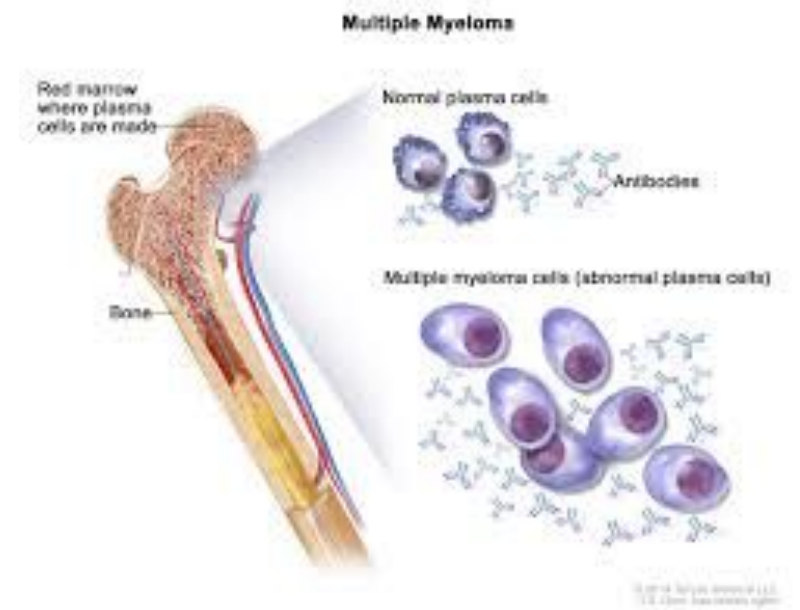
- * **Basic Structure:** Four polypeptide chains (2 heavy, 2 light) linked by disulfide bonds.
- * **Major Classes:** IgG, IgA, IgM, IgE, IgD.
- * **Key Dental Function:**
- * **Secretory IgA:** The primary immunoglobulin found in saliva and gingival crevicular fluid. It prevents bacterial adhesion to tooth surfaces and oral mucosa.

IMMUNOGLOBULIN ABNORMALITIES (DYSPROTEINEMIAS)

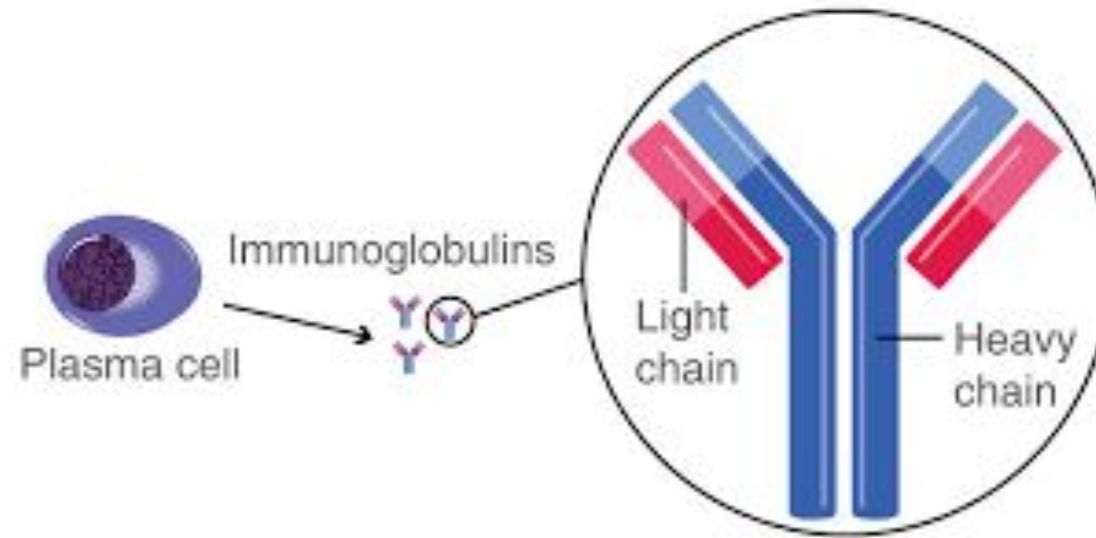
- * **Hypergammaglobulinemia:** Elevated Igs.
 - * **Polyclonal:** Chronic infections (e.g., chronic periodontitis, HIV, autoimmune diseases).
 - * **Monoclonal (Paraproteinemia):** Multiple Myeloma (cancer of plasma cells).
- * **Hypogammaglobulinemia:** Reduced Igs.
- * **Consequence:** Primary or secondary immunodeficiency leading to severe, recurrent infections.

MULTIPLE MYELOMA

- * **Biochemical Hallmark:** Bence-Jones proteins (light chains) in urine and "M-spike" on protein electrophoresis.
- * **Oral Manifestations:**
- * "Punched-out" radiolucent osteolytic lesions lesions.
- * Bone pain and loose teeth.
- * Macroglossia (enlarged tongue) due to amyloid deposition.



VARIANTS OF PCD



Multiple Myeloma

- * **What it is:** A malignant proliferation of plasma cells in the bone marrow.
- * **Key Feature:** Secretion of M-protein (monoclonal IgG or IgA) and Bence-Jones proteins (free light chains in urine).
- * **Result:** Multifocal destructive bone lesions ("punched-out" holes) and hypercalcemia.

Primary (Immunocyte-Associated) Amyloidosis

- * **What it is:** A PCD where the plasma cells produce excessive Light Chains (AL) that misfold.
- * **Key Feature:** These light chains deposit in tissues as insoluble fibrils (Amyloid).
- * **Result:** Systematic organ failure (heart and kidney), identifiable by Apple-green birefringence under polarized light.

Localized Plasmacytoma

- * **What it is**: A single, isolated tumor mass of neoplastic plasma cells.
- * **Key Feature**: Can be Soft Tissue (extra-medullary, usually in the respiratory tract) or Osseous (solitary bone lesion).
- * **Result**: Unlike Multiple Myeloma, it is localized and often curable with radiation.

Lymphoplasmacytic Lymphoma (Waldenström Macroglobulinemia)

- * **What it is**: A hybrid tumor of B-cells and plasma-like cells.
- * **Key Feature**: Secretion of massive amounts of IgM (a very large pentameric molecule).
- * **Result**: Hyperviscosity syndrome-the blood becomes thick like "sludge," causing vision and neurological issues.

Heavy Chain Disease

- * **What it is:** A rare disorder where only the Heavy Chain portion of the immunoglobulin is produced (no light chains).
- * **Key Feature:** Absence of light chains in the M-protein spike.
- * **Result:** Presents like a lymphoma, often involving the gut (Alpha-chain disease) or lymph nodes (Gamma-chain disease).

Monoclonal Gammopathy of Undetermined Significance (MGUS)

- * **What it is:** A "pre-malignant" or asymptomatic state.
- * **Key Feature:** M-protein is present in the blood, but at low levels ($< 3\text{g/dL}$), and there is no organ damage.
- * **Result:** Most cases stay stable, but about 1% per year progress to Multiple Myeloma.

SUMMARY & CLINICAL CORRELATION

- * **Diagnostic Tools:** Serum Protein Electrophoresis (SPEP) is the standard test to differentiate these abnormalities.
- * Chronic oral infection → Systemic elevation of APPs → Links to Cardiovascular disease.
- * Abnormal Ig patterns → Clue to underlying malignancy or severe systemic immune dysfunction



Thank You
Thank You
Thank You!!!!